0.1 Locomotion Evaluation

Evaluation metrics. Let N denote the number of measurement samples, $\Delta t = 0.02\,\mathrm{s}$ the control-step duration, and $d_a = 12$ the number of actuated joints. For body-frame longitudinal velocity, lateral velocity, and yaw rate, we define the tracking RMSE as

$$E_z = \sqrt{\frac{1}{N}\sum_{t=1}^{N}\left(z_t - z_t^{\mathrm{cmd}}\right)^2}, \qquad z \in \{v_x, v_y, \omega_z\}. \quad (1)$$

For reversal consistency, we compare forward and backward rollouts at the same speed magnitude and with conjugate gait commands. Let $\bar{\mu}_\theta^+(\phi)$ and $\bar{\mu}_\theta^-(\phi)$ denote cycle-averaged actor means at their decision phases, and let $\bar{\mathbf{c}}^+(\phi), \bar{\mathbf{c}}^-(\phi) \in [0,1]^4$ denote cycle-averaged contact probabilities. Using $M = 128$ phase samples, $\phi_m = m/M$, and the corrected alignment $\check{\phi}_m = (-\phi_m) \bmod 1$, we define

$$E_\mu = \sqrt{\frac{1}{Md_a}\sum_{m=0}^{M-1}\left\|\bar{\mu}_\theta^-(\phi_m) - \mathcal{T}_{\mathcal{A}}\left(\bar{\mu}_\theta^+(\check{\phi}_m)\right)\right\|_2^2}, \quad (2)$$

$$E_c = \frac{1}{4M}\sum_{m=0}^{M-1}\left\|\bar{\mathbf{c}}^-(\phi_m) - \bar{\mathbf{c}}^+(\check{\phi}_m)\right\|_1. \quad (3)$$

These metrics compare independently generated rollouts; they do not evaluate the actor on an exactly reversed observation history. The action transformation $\mathcal{T}_{\mathcal{A}}$ is the identity for the joint-position-offset actions used here.

We measure mechanical effort using mean absolute mechanical power and normalized torque-squared rate:

$$\begin{aligned} P_{\mathrm{abs}} &= \frac{1}{N}\sum_{t=1}^{N}\sum_{j=1}^{d_a}\left|\tau_{t,j}\dot{q}_{t,j}\right|, \\ U_\tau &= \frac{1}{N}\sum_{t=1}^{N}\sum_{j=1}^{d_a}\left(\frac{\tau_{t,j}}{\tau_j^{\mathrm{lim}}}\right)^2, \end{aligned} \quad (4)$$

where $\tau_{t,j}$ is applied torque, $\dot{q}_{t,j}$ is joint velocity, and τ_j^{lim} is the configured effort limit. P_{abs} is measured in watts and U_τ is dimensionless. These quantities describe actuator use rather than electrical energy consumption.

For each leg group $L \in \{\mathrm{F}, \mathrm{H}\}$, with joint indices $\mathcal{J}_L$, we compute normalized torque exposure and absolute work:

$$\begin{aligned} X_{\tau,L} &= \Delta t \sum_{t=1}^{N}\sum_{j\in\mathcal{J}_L}\left(\frac{\tau_{t,j}}{\tau_j^{\mathrm{lim}}}\right)^2, \\ X_{W,L} &= \Delta t \sum_{t=1}^{N}\sum_{j\in\mathcal{J}_L}\left|\tau_{t,j}\dot{q}_{t,j}\right|. \end{aligned} \quad (5)$$

Their front/hind imbalances are

$$I_k = 100\,\frac{|X_{k,\mathrm{F}} - X_{k,\mathrm{H}}|}{X_{k,\mathrm{F}} + X_{k,\mathrm{H}}}, \qquad k \in \{\tau, W\}, \quad (6)$$

with zero imbalance when both exposures are zero. We write I_τ and I_W for normalized torque and absolute-work imbalance, respectively. Each imbalance is calculated within a trial before averaging, so front-heavy and hind-heavy trials cannot cancel. Finally, the joint-target clipping percentage is

$$C_{\mathrm{clip}} = \frac{100}{Nd_a}\sum_{t=1}^{N}\sum_{j=1}^{d_a}\chi_{t,j}^{\mathrm{clip}}, \quad (7)$$

where $\chi_{t,j}^{\mathrm{clip}} = 1$ when a requested joint-position target is clipped and zero otherwise.

Comparison protocol. We evaluate the final policies after 20,000 training iterations over eight gait templates and six longitudinal commands, $v_x^{\mathrm{cmd}} \in \{-2, -1, -0.5, 0.5, 1, 2\}\,\mathrm{m/s}$, with zero lateral and yaw-rate commands. Each of the 48 conditions has 100 recorded trials. Rollouts include 5 s of settling and a planned 20 s measurement period, trimmed to complete gait cycles. We average trials within each condition, conditions within each gait family, and then the four families equally. Tables 1 and 3 report these means and the paired difference $\Delta = \mathrm{PPO\text{-}TRS} - \mathrm{PPO}$. Their 95% scenario-resampling intervals use 10,000 paired bootstrap replicates over conditions within each family. Each robot–method pair has one trained policy, so these intervals describe sensitivity to the evaluated conditions rather than uncertainty across training seeds. Table 2 reports the corrected-phase subset, using one representative trial per condition and 24 forward/backward pairs per policy.

Results. We first examine command tracking in Table 1. PPO-TRS reduces Go2's longitudinal and lateral velocity errors by 6.5% and 32.6%, respectively, with lateral tracking improving in 42 of the 48 gait–velocity conditions. The other configurations show clearer trade-offs. X1-Aligned achieves 8.8% lower mean longitudinal error, but its scenario-resampling interval for the difference spans zero; its lateral error increases by 19.6% and worsens across all eight gait means. X1-Opposed also has a 6.6% increase in mean longitudinal error. One possible explanation is that the regularizer discourages unnecessary differences between forward and backward responses, but can also oppose corrections needed for accurate tracking. Its objective penalizes action mismatch without directly considering the tracking quality of the paired actions, a limitation also discussed for symmetry losses by Mittal et al. [1]. The present comparisons do not isolate this mechanism.

Gait analysis requires distinguishing periodicity, time-reversal symmetry, and symmetry under leg exchange. Ding and Gan [2] show that these properties need not change together: their half-bounding branches lose the identified time-reversal symmetry, whereas some galloping solutions retain it despite lacking leg-permutation symmetries. This model-specific result motivates evaluating each realized gait rather than assuming that a more synchronized footfall pattern is more compatible with TRS. In our evaluation, trot and bound are paired with the same commanded template under reversal, while half-bound and gallop variants are paired with their conjugate variants; for example, gallop A is paired with gallop C. Reversal consistency therefore does not

Table 1: Command-tracking results. Differences and their 95% scenario-resampling intervals use the units of the corresponding metric. Negative differences indicate lower error under PPO-TRS. The last column counts conditions with lower error out of the 48 evaluated conditions.

Robot	Metric	PPO	PPO-TRS	Δ	95% scenario interval for Δ	Improved conditions
Go2	E_{v_x} [m/s]	0.2146	0.2007	−0.0140	[−0.0264, −0.0017]	32/48
	E_{v_y} [m/s]	0.0973	0.0655	−0.0318	[−0.0508, −0.0163]	42/48
	E_{ω_z} [rad/s]	0.0989	0.0995	+0.0006	[−0.0101, +0.0114]	24/48
X1-Opposed	E_{v_x} [m/s]	0.0912	0.0973	+0.0061	[−0.0081, +0.0211]	21/48
	E_{v_y} [m/s]	0.0559	0.0545	−0.0015	[−0.0100, +0.0062]	27/48
	E_{ω_z} [rad/s]	0.0799	0.0841	+0.0042	[−0.0049, +0.0136]	24/48
X1-Aligned	E_{v_x} [m/s]	0.0932	0.0850	−0.0082	[−0.0220, +0.0043]	27/48
	E_{v_y} [m/s]	0.0558	0.0668	+0.0109	[+0.0037, +0.0187]	19/48
	E_{ω_z} [rad/s]	0.0908	0.0877	−0.0031	[−0.0138, +0.0068]	27/48

Table 2: Behavioral reversal consistency on the corrected-phase subset of 24 forward/backward pairs per policy. Values are averaged with equal weights for the four gait families. Reductions are relative to PPO. These are descriptive sample results without training-seed uncertainty estimates.

Robot	Actor-mean RMSE E_μ [raw action]			Contact-pattern error E_c [fraction]		
	PPO	PPO-TRS	Reduction	PPO	PPO-TRS	Reduction
Go2	1.5130	1.3211	12.7%	0.1308	0.1007	23.0%
X1-Opposed	1.7799	1.7649	0.8%	0.1569	0.1491	5.0%
X1-Aligned	2.1538	1.9563	9.2%	0.1398	0.1349	3.5%

require preserving the same leading-leg variant. Table 2 shows that PPO-TRS reduces Go2's actor-mean consistency RMSE by 12.7% and contact-pattern error by 23.0% in the corrected-phase sample. X1-Aligned improves actor consistency by 9.2%, whereas X1-Opposed changes by only 0.8%. These measurements support improved behavioral consistency most clearly on Go2, but do not establish reversible dynamics or confirm that every commanded gait is executed correctly.

Finally, Table 3 distinguishes mechanical effort from its distribution. Go2's normalized torque imbalance decreases from 36.46% to 27.96%, improving across all eight gait means, although its mean mechanical power increases by 3.2%. Similarly, X1-Opposed uses 6.7% less normalized squared torque but 7.3% more mechanical power. Reduced actuator utilization or more balanced loading therefore does not necessarily imply lower energy consumption. The gait-level results reinforce this distinction: Go2's bound has better torque balance, but its work imbalance increases from 6.72% to 12.52%. Our time-reversal transformation does not exchange the front and hind legs, so equal workload is not imposed by the regularizer. Contact timing and pitching dynamics can also require different contributions from the two leg groups. Joint-target clipping decreases across all three configurations, and all three scenario-resampling intervals for its change lie below zero, indicating fewer out-of-range requests across the evaluated conditions.

Table 3: Mechanical effort, front/hind workload imbalance, and joint-target clipping. Differences and their 95% scenario-resampling intervals use watts for P_{abs}, dimensionless units for U_τ, and percentage points for I_τ, I_W, and C_{clip}. Lower imbalance means a more even load distribution; lower power alone does not establish better energy efficiency when tracking quality differs.

Robot	Metric	PPO	PPO-TRS	Δ	95% scenario interval for Δ
Go2	P_{abs} [W]	125.95	130.00	$+4.05$	$[-3.48, +12.93]$
	U_τ	0.4347	0.4545	$+0.0198$	$[+0.0040, +0.0356]$
	I_τ	36.46%	27.96%	-8.50	$[-11.49, -5.33]$
	I_W	12.69%	11.79%	-0.90	$[-4.44, +2.34]$
	C_{clip}	1.357%	0.921%	-0.436	$[-0.704, -0.184]$
X1-Opposed	P_{abs} [W]	185.20	198.74	$+13.54$	$[-6.14, +36.34]$
	U_τ	1.0496	0.9791	-0.0706	$[-0.1129, -0.0323]$
	I_τ	9.24%	10.78%	$+1.54$	$[-0.36, +3.52]$
	I_W	10.21%	9.20%	-1.01	$[-3.97, +1.83]$
	C_{clip}	1.348%	0.641%	-0.707	$[-1.157, -0.277]$
X1-Aligned	P_{abs} [W]	207.56	205.83	-1.73	$[-27.28, +25.21]$
	U_τ	1.0119	0.9812	-0.0307	$[-0.0821, +0.0201]$
	I_τ	12.72%	11.27%	-1.44	$[-4.78, +1.93]$
	I_W	14.01%	13.89%	-0.13	$[-4.94, +5.05]$
	C_{clip}	2.858%	1.724%	-1.134	$[-1.682, -0.582]$

References

[1] M. Mittal, N. Rudin, V. Klemm, A. Allshire, and M. Hutter, *Symmetry Considerations for Learning Task Symmetric Robot Policies*, 2024.

[2] J. Ding and Z. Gan, *Breaking Symmetries Leads to Diverse Quadrupedal Gaits*, 2024.